AI Literacy

POD 3 - Introduction to Generative AI

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You can't open up a social media stream or read the news or have a conversation with your co-workers without a mention of AI these days. In most cases, what people are talking about is Generative AI. You know Chat GPT and similar sort of chatbots that appear so humanlike and want to help you do anything from improving your resume. It seems it can do all, but what is Generative AI? As Suzanne showed in the previous part of this course, AI has been around for decades and mostly, it has been used for classification and prediction like assigning labels to photos and prediction like assigning labels to photos and predicting whether you'll watch a Netflix show recommended for you. Generative AI though is a form of AI that also takes training data, but then generates more of the same kind of training data. You can see this in the example on the screen where we take pictures of cats because AI people always love pictures of cats and we encode a representation of what a cat looks like and then we use a decoder to generate a new picture of a cat which is all the characteristics of what makes up a cat.

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So we have used training data to generate a model and then generative AI generates one of any number of pictures of cats based on the patterns and characteristics it experienced in the training data. If the training data we use is text, then we can build a model for the text and generate more of that text. So, if we were to take all all the poems of Seamus Heaney and encode them into a representation, which we would a language model, then we could generate Seamus Heaney lookalike poetry. About five or six years ago some big tech companies took this idea and decided to use not just small collections of poetry, but almost all of the text on the internet to encode patterns of works into a model or representation, which has become knows as a large language model or LLM. Interest in this idea soared in November 2022 when one of those companies, OpenAI coupled their LLM with a chatbot called the ‘ChatGPT’, released it to the public and the rest is history.

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The most common way to use an LLM, which everyone can do is via chat interface. So given some input and instruction or sentence known as a prompt, the LLM will generate the most likely sequence of words in response to that prompt, based on the patterns of text, it is learned from processing its huge amount of training data. The longer an interaction with an LLM goes on, i.e, the more interactions, the more credible will be the LLM’s response. This is shown in the animation on screen where a long, past history of interactions or a document introduced from fine tuning which we’ll learn about later on will yield incredible response. So, back to the LLM’s, just how big are they? Well, because they use so much training data and because they are used to predict so many words into the future, as part of some dialogue, they need to know about the likelihood or probabilities of all possible combinations of word sequences. So, they are huge! The diagram on screen shows the amount of training date used to train GPT4, which is already an old version of OpenAI’s LLM. If all the training data was printed into books, the line of books would be 650KM long and the time needed to turn all those books into a model, if it was done say on my Apple MacBook, it would take 2.9 millions years. But, of course OpenAI have tens and thousands of computers working on this, so it takes them three or four months only.

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And the LLM itself, well if it was printed out on sheets of paper and you know I have no idea why you would want to do this, but suppose you did, then, it would be twenty-five times the size of the Phoenix Park. So why does it take so long to turn that training data into a model, into an LLM? Well, I liken this to having a huge bag of coloured balls, like snooker balls, hundreds of thousands of them, each one representing a word and each one being a different colour to all the others. What we want to do in building an LLM is to arrange them on a very large snooker table so that, so that colours which are almost the same are beside each other. So every time you want to add another coloured ball on the table, you have to move others, rejig them, reorganise them to make room and you have to do this for every one of the hundreds of thousands of snooker balls. So that's what generative AI looks like for text and in a similar way, LLM's have also been used for creating images for videos and for mixtures of text and image and video, all bundled together. That is how we can use LLM's to generate, not just text, but also images and even videos.

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Finally, while OpenAI and their ChatGPT service is possibly the most well-known LLM, there are literally hundreds of others available to us at our fingertips. Microsoft are now using an LLM with a chat interface called Copilot, which also offers help in some of their 365 products like Word or Powerpoint. You may have seen this already because DCU subscribes to these services from Microsoft. Google have their own LLM called Gemini and if you have a personal gmail account, you can access Gemini through that. And a fourth example of an LLM besides ChatGPT, Copilot or Gemini is called Claude.ai, which comes from a company called Anthropic and it's important because it is not one of the established big tech companies and Claude is known to perform at the same level as the LLM's from those established companies. We will come back to see more of these in a later video.